Chrysanthi Kosyfaki

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About me

My research interests are in the area of graph analytics with a particular focus on a new class of networks called Temporal Interaction Networks (TINs.) In these networks, a quantity (flow) transfers from one vertice to another within a time window. TINs can model dynamic systems where entities exchange quantities, such as financial transactions, transportation flows, or communication data. Over the past few years, I have been developing efficient solutions to optimize classical problems—such as max-flow computation—by introducing the temporal dimension.
I have also explored the Text-to-SQL domain, focusing on addressing challenges related to text ambiguity in natural language database queries.
Currently, I am working on provenance tracking for streaming systems. As real-time analytics pipelines grow in complexity, understanding how outputs are derived from inputs becomes critical for debugging, auditing, and explainability. I am developing how-much provenance, a new model that efficiently counts input contributions across dataflow paths rather than enumerating individual tuples. This approach incorporates temporal window-based tracking and path compression techniques to scale to high-throughout streaming workloads, and supports queries that allow users to identify which inputs e.g., from which paths or time windows contributed to a given output.

Employment

Hong Kong University of Science and Technology, Hong Kong SAR *May 2025 – present*
Postdoctoral research fellow, CSE Department
- Working on projects related to spatiotemporal data management and data provenance on graphs

University of Hong Kong, Hong Kong SAR *Jan. 2025 – May 2025*
Part-time Lecturer, CS Department
- Taught a course at the master program of the department

University of Hong Kong, Hong Kong SAR *Aug. 2023 – Apr. 2025*
Postdoctoral research fellow, CS Department
- Worked on projects related to spatiotemporal data management and graph analytics

University of Hong Kong, SAR *Jun. 2022 – Oct 2022*
Research Assistant, CS Department
- Worked on developing spatiotemporal data management algorithms for a funding project with MTR

University of Ioannina, Greece *Oct 2020 – May 2023*
Software Developer, CSE Department
- Worked on a research funding project called SmartCityBus.

University of Ioannina, Greece *Apr. 2020 – Sept 2020*
Software Developer, CSE Department
- Worked on a research funding project called ProximIoT.

University of Hong Kong, SAR *Aug 2019 – Oct 2019*
Research Assistant, CS Department
- Worked on developing optimized algorithms for flow computation in temporal networks

University of Ioannina, Greece *Mar. 2019 – Jul 2019*
Software Developer, CSE Department
- Worked on a research funding project called Seek and Go.

University of Hong Kong, SAR *Aug. 2018 – Dec 2018*
Research Assistant, CS Department
- Worked on developing optimized algorithms for enumerating flow motifs in temporal networks

Education

Ionian University, Greece *Sept. 2013 – Sept 2017*
BS in Computer Science
- **Thesis**: Sentimental Analysis in Social Networks
- **Advisor**: Phivos Mylonas

University of Ioannina, Greece *Oct. 2017 – Feb 2019*
M.Sc in Computer Science
- **Thesis**: Flow Motifs in Interaction Networks
- **Advisor**: Nikos Mamoulis

University of Ioannina, Greece *Feb. 2019 – Apr 2023*
Ph.D in Computer Science
- **Thesis**: Flow Analytics in Large Graphs
- **Advisor**: Nikos Mamoulis

Academic Service

Organization Committees

Co-chair for TKDE posters @ICDE	2025-2026
Publicity chair for @MDM 2026	2026

Program Committees

PVLDB	2024-2027
ICDE	2025-2027
SIGSPATIAL	2025
VLDBJ	2024 - 2025
TKDE	2023-2025
PAKDD	2022-23

External Reviewer

ICDE	2019-2024
PVLDB	2019-2023
SIGMOD	2021-2023
EDBT	2018-2020
KDD	2019

Student Volunteer

PVLDB	2020
EDBT	2023

Awards

Christine Collet EDBT/ICDT Student Participation Award 2019 — 2019

Teaching Experience

The University of Hong Kong, SAR — *Spring 2025*
Course Title: Network Data Analytics

- *Instructor*

University of Ioannina, Greece — *Spring 2019-2023*
Course Title: Complex Data Management

- *Teaching Assistant*

University of Ioannina, Greece — *Spring 2018*
Course Title: Object Oriented Programming

- *Teaching Assistant*

University of Ioannina, Greece — *Fall 2017-2022*
Course Title: Introduction to Programming

- *Teaching Assistant*

Highlights

Temporal Interaction Networks (TINs) Temporal Interaction Networks (TINs) are a graph-based formalism for modeling time-varying data flows. Specifically, this type of network models dynamic systems where entities exchange quantities, such as financial transactions, transportation flows, or communication data. They can be used in a variety of problems; for computing the max-flow quantity from a source to a sink vertex to tracking the origin of a transferred quantity among the vertices (provenance).

BEACON - A Benchmark for Efficient and Accurate Counting of Subgraphs BEACON is a scalable and standardized dataset designed to evaluate ML-based techniques. It provides a unified dataset with ground truth subgraphs and different types of graphs (e.g., directed and undirected, weighted and unweighted).

Skills

Programming Languages
C, C++, Python

Environments
MATLAB, Octave, QGIS, Neo4j

Operating Systems
Windows, MacOS, Linux

Publications

Back, Forth, Across, and Through: Indexing Aggregate Temporal Provenance — **2027**
P.Simatis, **C.Kosyfaki**, X.Zhou and N.Mamoulis - **@VLDB** (*under submission*)

Enhancing Constrained Shortest Path Resilience with Approximate Skyline Board Index — **2026**
N.Shabuerjiang, Ziyi Liu, Lei Li, **C.Kosyfaki**, H. Huang, A. Zhou, W. Hua, and X.Zhou - **@VLDBJ** (*under review*)

Explainability in Data Management — **2026**
C.Kosyfaki and Y.Peng - **@VLDB** (*under review - Tutorial track*)

An (Updated) Overview of Data Provenance: Concepts, Challenges and Opportunities — **2026**
C. Kosyfaki, P. Groth - **@SIGMOD** (*as a tutorial*)

Multigranularity Spatiotemporal Flow Patterns — **2026**
C. Kosyfaki, N. Mamoulis, R. Cheng, B. Kao - under revision (Geoinformatica)

Toward Temporal Attribution Analytics in Dataflows **2026**
C. Kosyfaki, R. Zhang, N. Mamoulis, X. Zhou **@VLDB**

BEACON: A Benchmark for Efficient and Accurate Counting of Subgraphs **@ICDE** **2026**
X. Zhu, M. Najafi, **C. Kosyfaki**, L. VS. Lakshmanan, R. Cheng

Refine or Execute? A Cost-Based Framework for Socratic Query Refinement - **under submission** **2026**
R. Zhang, **C. Kosyfaki**, Sau Lai Yip, X. Zhou

Generalized Origin-Destination-Time Flow Patterns **@SSTD** **2025**
C.Kosyfaki, N. Mamoulis, R. Cheng, B.Kao

A Sampling-based Framework for Hypothesis Testing on Large Attributed Graphs **@PVLDB** **2024**
Y. Wang, **C. Kosyfaki**, S. Amer-Yahia, R. Cheng

SmartCityBus - A Platform for Smart Transportation Systems **@WSDM** **2023**
G. Bouloukakis, C. Zeginis, N. Papadakis, K. Magoutis, G. Christodoulou, **C. Kosyfaki**, K. Lampropoulos, N. Mamoulis

Spatiotemporal Flow Patterns **@arxiv** **2023**
C. Kosyfaki, N. Mamoulis, R. Cheng, Ben Kao

Provenance in Temporal Interaction Networks **@ICDE** **2022**
C.Kosyfaki and N. Mamoulis

Flow Provenance in Temporal Interaction Networks **@SIGMOD** (*short paper*) **2021**
C.Kosyfaki and N. Mamoulis

Flow Computation in Temporal Interaction Networks **@ICDE** **2021**
C.Kosyfaki, N. Mamoulis, E. Pitoura, P. Tsaparas

Flow Motifs in Interaction Networks **@EDBT** **2019**
C.Kosyfaki, N. Mamoulis, E. Pitoura, P. Tsaparas

Flow Motifs in Complex Networks **@HDMS** (*poster contribution*) **2018**
C.Kosyfaki

The Privacy Paradox in the Context of Online Health Data Disclosure by Users **@EMCIS** **2017**
C.Kosyfaki, N. Angelova, A. Tsohou, E. Mangos

Student Supervision

PhD Students **HKUST**
- **TBA**

PhD Students **HKU**
- **Carrie Wang** - *working on hypothesis testing on graphs*
- **Xiangju Zhu** - *working on subgraph counting problems*
- **Matin Najafi** - *worked on subgraph counting problems, now a researcher at Huawei, Hong Kong*

Bachelor Students **UoI**
- **Ioanna Papayianni (2020)** - *Thesis: Developing efficient algorithms for analyzing flow patterns in large networks*
- **Dimitris Zervas (2021)** - *Thesis: Detecting the origin of transactions in the Bitcoin Network*
- **Sotiria Kastana (2021)** - *Thesis: Design and development of a synthetic indoor movement generator*
- **Vasileios Georgoulas (2023)** - *Thesis: A web application for passenger movement with public transport*